Lucas Toñanez

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EDUCATION

University of Central Florida **Orlando, FL**

Bachelor of Science in Computer Science *Expected May 2027*

- **GPA:** 3.4/4.0 | Bright Futures Scholarship | Millican Gold Merit Award
- **Coursework:** Artificial Intelligence, Algorithms for Machine Learning, Data Structures & Algorithms, Software Development

PUBLICATION

Conditional Multi-Event Temporal Grounding in Long-Form Video arXiv:2606.15320 (2026)

*Yuanhao Zou, Arthad Kulkarni, **Lucas Toñanez**, et al. — Third author; Computer Vision (cs.CV)*

TECHNICAL EXPERIENCE

Tecalemit Inc. — Software Engineering Intern, AI & Automation **Sep 2024–2025**

- Built and maintained **internal software tools and REST API integrations** supporting company workflows, product information, and structured data exchange.
- Developed AI-assisted customer-service and sales workflows using OpenAI APIs, backend services, structured product context, and automated email handoffs.
- Tested, debugged, and documented software across local and deployed environments using Git, Postman, Linux, and server tooling to resolve API and deployment issues.

UCF Institute for Artificial Intelligence — Undergraduate Research Assistant **Dec 2025–Present**

- Integrated independently developed components into an **end-to-end Python multimodal pipeline** for query planning, video processing, model inference, temporal localization, and structured prediction.
- Built data and evaluation workflows spanning **600+ long-form videos** with JSONL extraction, checkpoint/resume, runtime/error tracking, consistency validation, and reproduction of Qwen, VideoMind, and DoraemonGPT baselines.
- Reduced scene-detection workload from **307 to 91 candidate segments (~70%)** in a profiled run through query-window clipping/caching; also implemented batched GPU inference and CUDA resource management.

UCF Institute for Artificial Intelligence — Undergraduate Researcher — VLA / Robotics **Jun 2026–Present**

- Supported **real-robot deployment and rollout recording** using OpenPI, LeRobot, and ROS2; collected success/failure feedback and traced data flow across evaluation and model-update stages.

UCF Institute for Artificial Intelligence — Undergraduate Researcher — Video Annotation **Sep 2026–Present**

- Built a Qwen3-VL-8B Annotator/Corrector video-annotation pipeline with a Qwen2.5-VL baseline, frozen-draft evaluation, and 100+ CPU contract tests; designed a frame-cited boundary-evidence protocol for temporal corrections.

SELECTED PROJECTS

Mindful AI — Backend Developer (UCF Senior Design) **Supabase, PostgreSQL**

- Built an authenticated Supabase Edge Function (Deno/TypeScript) for AI-generated mindfulness responses, connected it to the team's LLM provider adapter, and persisted generated intentions idempotently behind PostgreSQL row-level security.
- Designed migrations and RLS policies for per-beat progress, baseline-response, and Week 3 draft persistence; wrote endpoint and database integration tests against local Supabase, an ERD/data dictionary, and a backend operations guide.

Video Game Store Web Application **MERN, Linux**

- Built and deployed a full-stack store with authentication, protected routes, REST APIs, MongoDB models, and React/TypeScript interfaces on DigitalOcean using PM2 and Nginx.

Gasoline Price Movement Classification (CAP 5610) **PyTorch, NumPy**

- Built a leakage-safe pipeline over 142,818 French station–fuel sequences and compared MLP, 1D CNN, LSTM, and Transformer models via an 80-trial search with 3-seed reruns; Transformer reached 0.666 validation macro-F1 vs. 0.635 for logistic regression.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Development: React, Node.js, Express, MongoDB, Supabase, PostgreSQL, Deno, FastAPI, Flask, REST APIs, Git/GitHub, Postman
Systems & Data: Linux, CUDA, JSON/JSONL, FFmpeg, OpenCV, ROS2, PM2, Nginx, DigitalOcean
AI/ML & Robotics: PyTorch, scikit-learn, Hugging Face, Qwen, DINOv2, RAFT, OpenPI, LeRobot